1. Thermal Storage & Loss

$$C_{\mathbf{c}} \frac{\partial T_{\mathbf{c}}}{\partial t} = -K(T_{\mathbf{c}} - T_{\mathbf{l}})$$

Carrier Subsystem (T_c) → **Phonon Loss** $-K(T_c - T_l)$ → **Lattice Subsystem (T_l)**

2. Optical Excitation (Laser)

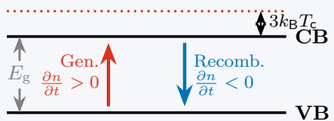
$$+(\alpha + \beta I + \Theta n)\, I$$

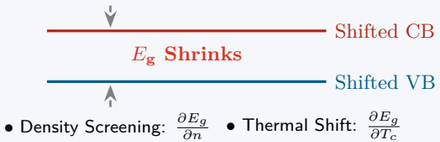
Laser I

- → **Single-Photon:** αI
- → **Two-Photon:** βI^2
- → **Free-Carrier:** $\Theta n I$

3. Carrier Generation & Recombination

$$-\frac{\partial n}{\partial t}\left(E_{\mathbf{g}} + 3k_{\mathbf{B}}T_{\mathbf{c}}\right)$$



4. Bandgap Renormalization (BGR)

$$-n\left(\frac{\partial E_{\mathbf{g}}}{\partial n}\frac{\partial n}{\partial t} + \frac{\partial E_{\mathbf{g}}}{\partial T_{\mathbf{c}}}\frac{\partial T_{\mathbf{c}}}{\partial t}\right)$$